

Pakistan studies



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Chapter 03



In the name of Allah, the Most Merciful, the Most Kind

Geography of Pakistan

1 – Climatic Regions (Key Terms):

- **Climate** – The average weather of a place over a long time
- Weather** – The daily condition of air (hot, cold, rainy, etc.)
- Monsoon Winds** – Seasonal winds that bring rain
- Rainfall** – The amount of rain received
- Temperature** – The level of heat or cold
- Humidity** – Moisture in the air
- Arid Region** – Very dry area with little rain
- Semi-Arid Region** – Area with small amount of rain
- Drought** – A long period without rain
- Heat Wave** – Many extremely hot days
- Western Disturbances** – Winter rain-bringing winds in Pakistan

- **2 – Hydrological Environment (Key Terms):**

- **Hydrology** – Study of water

Hydrological Cycle – Continuous movement of water on Earth

Evaporation – Water changing into vapor by heat

Condensation – Vapor turning into water drops

Precipitation – Rain, snow, or hail falling from clouds

Runoff – Water flowing on land surface

Infiltration – Water soaking into the ground

Groundwater – Water stored under the earth

Glacier – Large mass of ice

River Basin – Area drained by a river

Reservoir – Stored water (usually behind a dam)

3 . Physiography of Pakistan:

- **Physiography** – Physical features of land
- Northern Highlands** – High mountain region of Pakistan
- Balochistan Plateau** – High flat dry land
- Indus Plain** – Fertile flat land along River Indus
- Alluvial Soil** – Soil made by river deposits
- Desert** – Dry sandy region
- Mountain Range** – Long line of mountains
- Valley** – Low land between mountains
- Coastal Belt** – Area along the sea
- Plateau** – Raised flat land
- Plains** – Flat low land

Physiography of Pakistan

1. Meaning of Physiography

- Physiography means the **physical features of the land** of a country.
It includes **mountains, plateaus, plains, deserts, rivers, and coastal areas.**
- Pakistan has **very diverse physical features**, from high mountains in the north to plains and deserts in the south.

- **2. Major Physiographic Divisions of Pakistan**
- Pakistan is divided into **six main physiographic regions:**
- **A. Northern Mountains**
- These mountains are located in **northern Pakistan** and are among the **highest in the world.**
- **Main Mountain Ranges**
- **Himalayas**
- **Karakoram**
- **Hindu Kush**

- **1. Himalayas**
- **Location**
- Located in **northeast Pakistan**
- Spread over **Azad Kashmir and northern Khyber Pakhtunkhwa**
- **Main Features**
- Very **high and young fold mountains**
- Heavy snowfall
- Source of many rivers
- Rich forests

Important Peaks:

- **Nanga Parbat (8,126 m)** – *Killer Mountain*
- **Rakaposhi (7,788 m)**
- **2. Karakoram Range:**
 - **Location**
 - Lies in **Gilgit-Baltistan**
 - Between **Himalayas and Hindu Kush**
 - **Main Features:**
 - Home of the **highest peaks of Pakistan**
 - Very cold climate
 - Large glaciers (Siachen, Baltoro)
 - Rugged and rocky terrain

- **Important Peaks:**
- **K2 (8,611 m)** – *Second highest peak in the world*
- **Gasherbrum I (8,080 m)**
- **Gasherbrum II (8,035 m)**
- **Broad Peak (8,051 m)**
- **Importance:**
Major source of **glaciers**, feeding the **Indus River**.

- **3. Hindu Kush Range:**
- **Location:**
- Located in **north-west Pakistan**
- Extends from **Afghanistan into Chitral and Dir**
- **Main Features**
- Steep slopes
- Narrow valleys
- Cold winters
- Less snowfall than Karakoram

- **Important Peaks:**
- **Tirich Mir (7,708 m)** – *Highest peak of Hindu Kush*
- **Noshaq (7,492 m)** (*on Pak-Afghan border*)
- **Importance:**
Acts as a **natural barrier** and affects **regional climate**.

- **Western Mountains:**

- These mountains lie in **western Pakistan**, mainly in **Balochistan and Khyber Pakhtunkhwa**.

- **Main Ranges**

- **Sulaiman Range**

- **Kirthar Range**

- **Makran Range**

- **Features:**

- Lower than northern mountains
- Dry and rocky
- Act as barriers to monsoon winds

- **Importance:**

They affect climate and limit rainfall in western areas.

- **Main Ranges & Important Peaks**
- **1 Sulaiman Range:**
- **Location:** Baluchistan–KPK border
- **Important Peak:**
 - **Takht-e-Sulaiman (3,487 m)** → *Highest peak of Sulaiman Range*
- **Importance:**
 - Natural barrier between **Indus Plain and Baluchistan Plateau**
 - Historic invasion routes (e.g., Bolan Pass nearby)

- **Kirthar Range:**
- **Location:** Sindh–Baluchistan border
- **Highest Peak:**
 - **Zardak Peak ($\approx 2,300$ m)**
- **Importance:**
 - Separates **Lower Indus Plain** from **Baluchistan Plateau**
- Mostly dry and barren
- **Key Features:**
- **Climatic Importance :**
- Act as **barriers to monsoon winds**
- Cause **low rainfall** in western Pakistan
- Responsible for **arid climate** of Baluchistan

- **Makran Range:**
- **Location:** Southern Baluchistan (along Makran coast)
- **Important Peaks:**
 - No very high peaks (comparatively low range)
- **Importance:**
 - Extends parallel to **Arabian Sea**
- **“Earthquakes are common here due to movement of tectonic plates.”**

- **A) Arid (Desert) Climate Areas:**

- Thar Desert

- Cholistan

- Thal

- Kharan Desert

- **Key Features:**

- **Very low rainfall**

- **Hot summers, cold winters**

- **Sparse vegetation (opposite):** *Very few plants grow*

- **B) Semi-Arid Climate Areas:**
- Central Punjab
- Parts of KPK plains
- **Key Features:**
- Moderate rainfall
- Hot summers, mild winters
- Suitable for agriculture with irrigation

- **C) Sub-Humid / Humid Climate Areas:**
- Upper Punjab
- Northern KPK (Hazara region)
- **Key Features:**
- Higher rainfall
- Humid summers
- Dense vegetation

- **D) Highland Climate Areas:**

- Northern Areas
- Gilgit-Baltistan
- Northern KPK

Features:

- Cold climate
- Heavy snowfall
- Short summers

- **Northern Khyber Pakhtunkhwa (KPK) consists mainly of(Hazara Division):**
- **Major Areas of Northern KPK (Hazara Region):**
- **Abbottabad**
- **Mansehra**
- **Haripur**
- **Battagram**
- **Kohistan**

Hydrological environment

Meaning:

- The **hydrological environment** refers to all the water found on Earth and the natural process through which water moves in different forms.

It includes:

- Rivers
- Lakes
- Seas & oceans
- Rain
- Snow
- Underground water
- In simple words:
It is the **Earth's water system**.



- **Hydrological Cycle (Water Cycle):**
- Water continuously moves in a cycle called the **hydrological cycle**:
- **1. Evaporation:**
- Sun heats water from rivers, seas, and lakes
➡ Water changes into vapor (gas)
- **2. Condensation**
- Water vapor cools in the sky
➡ Forms clouds

- **3. Precipitation**

- Clouds release water as:

- Rain

- Snow

- Hail

- **4. Collection:**

- Water flows into rivers, lakes, and oceans
Some goes underground

Main Parts of Hydrological Environment

- **✓ Surface Water**
 - Water on the land surface:
 - Rivers
 - Lakes
 - Ponds
- **✓ Groundwater**
 - Water stored under the earth in soil and rocks
Used through:
 - Wells
 - Tube wells

- **Importance of Hydrological Environment**

- Provides drinking water
- Helps in agriculture (irrigation)
- Supports plants, animals & humans
- Controls climate & rainfall
- Produces electricity (hydropower)